

Ivan Lykov

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Education

MIPT, Masters in Computer Science – Moscow Sept 2024 – May 2026

- Thesis: Hybrid Hash Join implementation in YDB

YSDA, Computer Science – Moscow Sept 2024 – May 2026

NUST MISIS, Bachelors in Applied Mathematics – Moscow Sept 2020 – May 2024

Experience

Database SWE, YDB, Yandex – Moscow December 2025 – August 2026

- Developed a new index type to reduce query latency, achieving a 30% latency improvement on the ClickBench benchmark and 2× walltime speedup on client's timeseries queries
 - Studied index design in the most popular OLAP systems (ClickHouse, DuckDB, Snowflake), and built one similar to ClickHouse's
- Increased observability by adding per-query statistics to the developer UI

Database SWE Intern, YDB, Yandex – Moscow August 2025 – November 2025

- Implemented a new hash join operator:
 - Studied join implementations in ClickHouse, DuckDB, TiDB, and Postgres before designing the operator
 - 2× faster than its predecessor at the operator level without spilling, 6× faster with spilling to disk
 - Yields 10% end-to-end query latency improvement on the TPC-H s10k benchmark (sum of all 22 query latencies)
 - Consumes 2–10× less RAM
 - Eliminates OOM errors caused by unbounded memory growth in the predecessor

Misc

All my contributions to YDB are available in the YDB repository [here](#).

I also enjoy contributing to large open-source C++ projects, e.g. [Apache Arrow](#) and [LLVM/Clang](#).